

✓ AI Meeting Notes Summarizer

Project 2 — AI Automation Portfolio

This project uses Python, Prompt Engineering, and the OpenAI API to convert unstructured meeting notes into clear and organized business summaries.

The AI identifies:

- Main discussion points
- Key decisions
- Action items
- Responsible people
- Deadlines
- Follow-up tasks

Project Goal

The goal of this project is to automate the process of reviewing meeting notes and converting them into a structured summary that helps teams quickly understand what was discussed, what decisions were made, and what actions need to be completed.

✓ Step 2: Prompt Design and Core Logic

In this section, we create sample meeting notes and design a structured prompt that instructs the AI to identify and organize important meeting information.

```
# Sample unstructured meeting notes
```

```
meeting_notes = """
```

```
The marketing team met on August 1 to discuss the upcoming product launch.
```

```
Sameer explained that the social media campaign is approximately 70% complete.  
He will finish the remaining content and submit it for review by August 8.
```

```
Hassan reported that the advertising budget is still being reviewed.  
He will share the final budget with the team by August 6.
```

```
The team agreed to launch the new product on September 15.
```

```
The website landing page needs additional improvements before the launch.  
Ali will coordinate with the web development team and provide an update during  
the next meeting.
```

```
The next marketing meeting is scheduled for August 10.  
"""
```

```
# Create a structured prompt for meeting summarization
```

```
prompt = f"""
```

```
You are an expert business meeting assistant.
```

```
Your task is to analyze the meeting notes below and create a clear,  
accurate, and well-organized meeting summary.
```

```
Meeting Notes:
```

```
{meeting_notes}
```

```
Follow these instructions:
```

1. Write a short overview of the meeting.
2. Identify the main discussion points.
3. List all important decisions.
4. Extract every action item.
5. Include the responsible person for each action item when available.
6. Include deadlines when they are mentioned.
7. Identify any follow-up tasks.
8. Do not invent information that is not included in the meeting notes.
9. If information is missing, write "Not specified."
10. Use clear headings and bullet points.
11. Keep the summary concise and professional.

Use this output structure:

```
## Meeting Overview
```

```
## Main Discussion Points
```

```
## Key Decisions
```

```
## Action Items
```

```
## Follow-Up Items
```

```
## Next Meeting  
"""
```

```
print(prompt)
```

You are an expert business meeting assistant.

Your task is to analyze the meeting notes below and create a clear, accurate, and well-organized meeting summary.

Meeting Notes:

The marketing team met on August 1 to discuss the upcoming product launch.

Sameer explained that the social media campaign is approximately 70% complete. He will finish the remaining content and submit it for review by August 8.

Hassan reported that the advertising budget is still being reviewed. He will share the final budget with the team by August 6.

The team agreed to launch the new product on September 15.

The website landing page needs additional improvements before the launch. Ali will coordinate with the web development team and provide an update during the next meeting.

The next marketing meeting is scheduled for August 10.

Follow these instructions:

1. Write a short overview of the meeting.
2. Identify the main discussion points.
3. List all important decisions.
4. Extract every action item.
5. Include the responsible person for each action item when available.
6. Include deadlines when they are mentioned.
7. Identify any follow-up tasks.
8. Do not invent information that is not included in the meeting notes.
9. If information is missing, write "Not specified."
10. Use clear headings and bullet points.
11. Keep the summary concise and professional.

Use this output structure:

```
## Meeting Overview
```

```
## Main Discussion Points
```

```
## Key Decisions
```

```
## Action Items  
  
## Follow-Up Items  
  
## Next Meeting
```

✓ Step 2.2: AI-Powered Meeting Summary

In this section, the structured meeting prompt is sent to the OpenAI API. The AI analyzes the meeting notes and generates a clear, organized, and professional summary.

```
!pip install -q openai
```

```
from google.colab import userdata  
  
api_key = userdata.get("OPENAI_API_KEY")
```

```
from openai import OpenAI  
  
client = OpenAI(api_key=api_key)
```

```
response = client.responses.create(  
    model="gpt-5-mini",  
    input=prompt  
)  
  
meeting_summary = response.output_text  
  
print(meeting_summary)
```

```
print(meeting_summary)
```

✓ Testing the output of the last code with my API Key got the following result

Meeting Overview

Main Discussion Points

- Product launch preparation
- Social media campaign progress
- Advertising budget review
- Website landing-page improvements

Key Decisions

- The product will launch on September 15.

Action Items

- Sameer: Complete and submit the remaining social media content by August 8.
- Hassan: Share the final advertising budget by August 6.
- Ali: Coordinate with the web development team and provide an update.

Follow-Up Items

- Review website landing-page progress.
- Review the final advertising budget.
- Review the completed social media content.

Next Meeting

- August 10

Note:

The next exact output may be different. That is normal because AI responses can vary.

Quality Check

Review the output and confirm:

- The meeting overview is accurate.
- The main discussion points are included.
- The September 15 launch decision is included.
- Sameer's task and August 8 deadline are included.
- Hassan's task and August 6 deadline are included.
- Ali's responsibility is included.
- The next meeting date is included.
- No information was invented.

✓ Test a New Meeting Scenario

We'll use a product development meeting.

by replacing the Sample Meeting Notes

let's go back to the existing meeting_notes code cell and replace the previous notes with:

```
# Sample product development meeting notes

meeting_notes = """
The product development team met on August 2 to review progress on the
new mobile application.

Maria reported that the user registration feature is complete and has
successfully passed initial testing.

Usman explained that the payment integration is still in development.
He expects to complete the first version by August 12.

The team discussed adding a password reset feature. After reviewing
customer feedback, the team agreed that this feature should be included
before the first public release.

Ayesha will create the password reset feature requirements and share
them with the development team by August 7.

The team also discussed the application testing schedule.
Final testing will begin on August 15.

The project manager requested that all major development tasks be
completed before final testing begins.

The next product development meeting is scheduled for August 14.
"""
```

✓ Generate the New Summary

Run the existing AI-generation cell again:

```
response = client.responses.create(
    model="gpt-5-mini",
    input=prompt
)
```

```
meeting_summary = response.output_text  
  
print(meeting_summary)
```

The AI generated summary includes:

- Mobile application development progress
- Completion of the user registration feature
- Payment integration development
- The decision to add password reset functionality
- Maria's completed work
- Usman's August 12 target
- Ayesha's August 7 task
- Final testing beginning on August 15
- The next meeting on August 14

Evaluate the Output

Review the generated summary using this checklist:

Evaluation Area Expected Result

Meeting overview Accurately describes the mobile application project

Discussion points Includes registration, payments, password reset, and testing

Key decisions Includes the decision to add password reset

Action items Includes Ayesha's requirements task

Deadlines Includes August 7, August 12, and August 15

Next meeting Includes August 14

Accuracy Does not invent information

Organization Uses clear headings and bullet points

We can write **Pass** or Needs **Improvement** next to each item.

✓ Refine the Prompt

Now we'll improve the prompt so the AI handles action items and missing information more consistently.

Replacing the current prompt with the following refined code version:

```
# Refined prompt for structured meeting summarization  
  
prompt = f"""  
You are an expert business meeting analyst.  
  
Your task is to analyze the meeting notes and create an accurate,  
concise, professional, and action-oriented meeting summary.  
  
Meeting Notes:  
{meeting_notes}  
  
Follow these rules carefully:  
  
1. Use only information provided in the meeting notes.  
2. Do not invent names, dates, decisions, deadlines, or responsibilities.  
3. Clearly separate completed work from future action items.  
4. Extract every action item mentioned in the notes.
```

```

5. For each action item, identify:
    - Task
    - Responsible person
    - Deadline
6. If a responsible person or deadline is not mentioned, write
   "Not specified."
7. Include important project updates and completed work.
8. Include all confirmed decisions.
9. Include important follow-up items.
10. Keep the summary concise and easy to scan.
11. Use professional language.
12. Use bullet points where appropriate.

```

Use exactly this output structure:

```
## Meeting Overview
```

```
## Key Updates
```

```
## Main Discussion Points
```

```
## Key Decisions
```

```
## Action Items
```

```

| Task | Responsible Person | Deadline |
|---|---|---|

```

```
## Follow-Up Items
```

```

## Next Meeting
"""

```

Why This Prompt Is Better

The original prompt asked the AI to identify important information.

The refined prompt adds stronger instructions for:

- Separating completed work from future tasks
- Extracting action items consistently
- Organizing responsibilities and deadlines in a table
- Avoiding invented information
- Handling missing information clearly
- Producing an action-oriented business summary

▼ Step 3.2: Interactive Meeting Notes Summarizer

This section allows users to paste their own meeting notes and generate a structured AI-powered meeting summary without manually editing the Python code.

```

# Collect meeting notes from the user

print("Paste your meeting notes below.")
print("When you finish, press Enter on an empty line.")

meeting_lines = []

while True:
    line = input()

    if line == "":
        break

    meeting_lines.append(line)

```

```
meeting_notes = "\n".join(meeting_lines)

print("\nMeeting notes received successfully.")
```

```
# as a user you can put the following line of the variable one by one

samplenotes_one_line_at_a_time = ""
The sales team met to review monthly performance.

Amina reported that sales increased by 12 percent compared with the previous month.

The team agreed to focus more on repeat customers during the next campaign.

Bilal will prepare a customer retention plan by August 9.

The sales manager will review the plan and provide feedback by August 12.

The next sales meeting is scheduled for August 16.
After the final line, press Enter one additional time on an empty line
"""
# the resut will disply like this: "Meeting notes received successfully"
```

The summary should include:

- Sales performance increased by 12%
- A decision to focus on repeat customers
- Bilal's customer retention plan
- August 9 deadline
- Manager review by August 12
- Next meeting on August 16

Step 3.3: Final Testing

Let's test the summarizer with one more realistic business scenario.

Part A

— Test a Customer Support Meeting

Run all interactive meeting-notes input cell again.

paste the following notes one line at a time:

The customer support team met to review customer satisfaction and response times.

Hassan reported that the average customer response time improved from 10 hours to 6 hours during the previous month.

The team discussed several customer complaints related to delayed order deliveries.

The operations manager confirmed that the delivery process will be reviewed.

The team agreed to create a new customer support FAQ page to reduce repeated questions.

Fatima will collect the most common customer questions and prepare the first FAQ draft by August 11.

Hassan will analyze the delivery-related complaints and share a report by August 13.

The customer support manager will review both documents and provide feedback during the next meeting.

The next customer support meeting is scheduled for August 18.

After the final line, press Enter one additional time on an empty line.

You should see:

Meeting notes received successfully.

Last Result — Generate the Summary

Run:

- The refined prompt cell
- The AI-generation cell

The summary should include:

Key Updates

- Average response time improved from 10 hours to 6 hours.

Key Decisions

- Create a customer support FAQ page.

Action Items _ Fatima: Collect common customer questions and prepare the first FAQ draft by August 11.

- Hassan: Analyze delivery-related complaints and share a report by August 13.
- Customer Support Manager: Review both documents and provide feedback during the next meeting.

Next Meeting

- August 18

Evaluate the Final Output

Use this checklist:

- The response-time improvement is accurate.
- The delivery complaints are included.
- The FAQ-page decision is included.
- Fatima's task is included.
- August 11 is included.
- Hassan's task is included.
- August 13 is included.
- The manager's review task is included.
- The next meeting date is included.
- No information was invented. The summary is clear and professional.

AI Meeting Notes Summarizer

Project Goal

Step 2: Prompt Design and Core Logic

- Sample meeting notes
- Structured prompt

Step 2.2: AI-Powered Meeting Summary

- OpenAI API connection
- First generated summary

Step 3.1: Testing and Prompt Refinement

- Product development meeting
- Refined prompt

Step 3.2: Interactive Meeting Notes Summarizer

- User input workflow

Sample Output 1: Product Development Meeting

- Generated summary

Sample Output 2: Customer Support Meeting

- Generated summary